Introduction to Computers

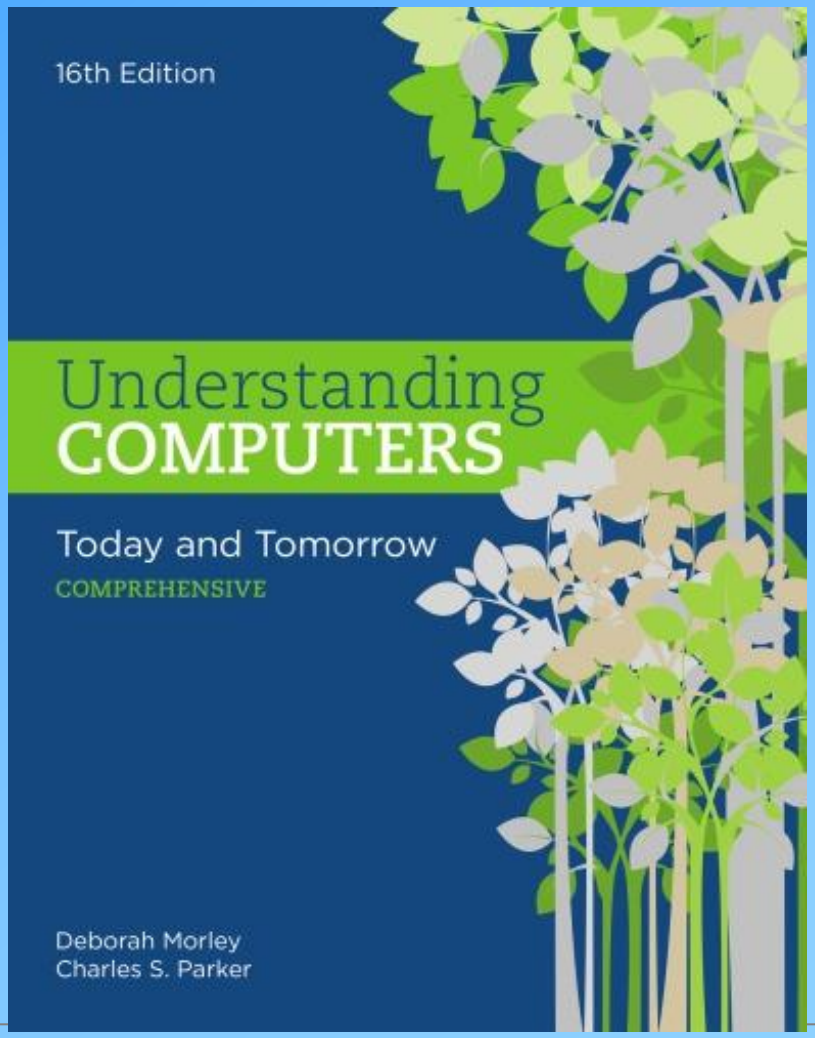
Course Aims and Objectives

Upon completion of this course, students should be able to:

1. Understand and state the importance of computer literacy.
2. Identify and explain what computers are and how they work, including the computer hardware components and their specifications and types.
3. Understand and use the binary system.
4. Understand system and application software with examples.
5. Understand the main issues of data communications and computer networks.
6. Describe the impact that computers are having on our society

Text Book

Understanding Computers – Today and Tomorrow – 16th Edition (2016)

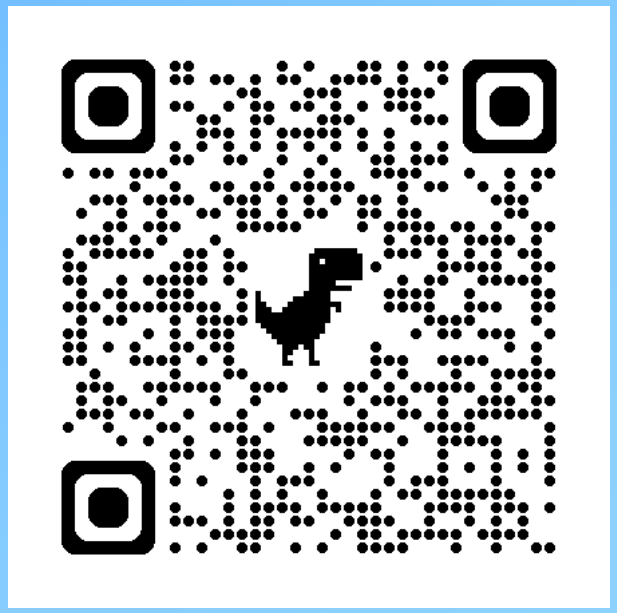


General Regulations

Students should only attend lectures and assessments in their own classes, labs and sections otherwise marks will not be considered and this might cause student failure

Shared folder for the course (Slides, Reference, etc…):

https://drive.google.com/drive/folders/1CY5JRkTiiWTVt_BDvXKjBBTxMPJ_nIPt?usp=sharing



Lecture 1

Introduction to the World of Computers

Reference :Understanding Computers
Chapter 1

Learning Objectives

- Explain why it is essential to learn about computers today and discuss several ways computers are integrated into our business and personal lives.
- Define a computer and describe its primary operations.
- List some important milestones in computer evolution.
- Identify the major parts of a personal computer, including input, processing, output, storage, and communications hardware.
- Define software and understand how it is used to instruct the computer what to do.
- List the five basic types of computers, giving at least one example of each type of computer and stating what that computer might be used for.

Computers in Your Life

- Why learn about computers?
- Why do you need basic computer literacy?
- List all the ways we use computers in our lives..
    - Looking up information and news
    - Exchanging e-mail
    - …
-

Computers Then and Now

Computers Then and Now

- The computer as we know it is a fairly recent invention
- **Precomputers** and early computers (before 1946)
  - Abacus, slide rule, mechanical calculator
  - Punch Card Tabulating Machine
- The history of computers is often referred to in terms of generations
- Each new generation is characterized by a major technological development





Computers Then and Now

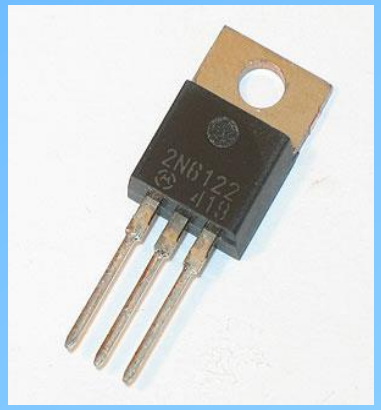
- First-generation computers (1946-1957)
  - Enormous and powered by vacuum tubes
  - Used a great deal of electricity, and generated a lot of heat



Computers Then and Now

- Second-generation computers (1958-1963)
  - Used transistors
  - Computers were smaller, more powerful, cheaper, more energy-efficient, and more reliable
  - Punch cards and magnetic tape were used to input and store data

Transistor



Computers Then and Now

- Third-generation computers (1964-1970)
  - Used integrated circuits (ICs)
  - Keyboards and monitors introduced
- Fourth-generation computers (1971-present)
  - Invention of Microcomputers in the early 80s
  - Use **microprocessors**
  - IBM PC, Apple Macintosh
  - Use keyboards, mice, monitors, and printers
  - Use magnetic disks, flash memory, and optical disks for storage
  - Computer networks, wireless technologies, Internet introduced

I.C.



Computers Then and Now

- Fifth-generation (now and the future)
  - Infancy stage
  - No precise classification
  - May be based on artificial intelligence (AI)
  - Voice and gesture
  - Based on optical computers and utilize nanotechnology

Evolution of Computing



1950

No Computers

1960s

Very expensive computers for large companies. Computer professionals ran the show.

mid-1970s

PCs introduced!

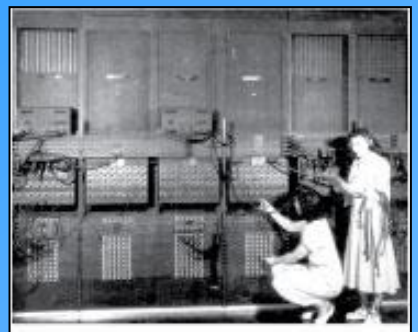
2003

Powerful PCs on every desktop. Homes networked.

Computers Then and Now



PRECOMPUTERS AND EARLY COMPUTERS
Dr. Herman Hollerith's Punch Card Tabulating Machine and Sorter is an example of an early computing device. It was used to process the 1890 U.S. Census data.



FIRST-GENERATION COMPUTERS
First-generation computers, such as ENIAC shown here, were large and bulky, used vacuum tubes, and had to be physically wired and reset to run programs.



SECOND-GENERATION COMPUTERS
Second-generation computers, such as the IBM 1401 mainframe shown here, used transistors instead of vacuum tubes so they were smaller, faster, and more reliable than first-generation computers.



THIRD-GENERATION COMPUTERS
Third-generation computers used integrated circuits which allowed the introduction of smaller computers, such as the IBM System/360 mainframe shown here.

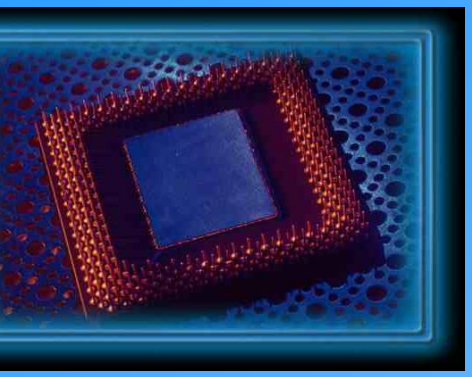


FOURTH-GENERATION COMPUTERS
Fourth-generation computers, such as the original IBM PC shown here, are based on microprocessors. Most of today's computers fall into this category.

FIGURE 1-7
A brief look at computer generations.

How Fast Is 1 GHz?

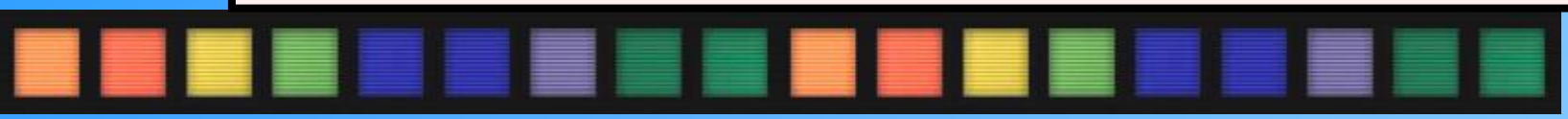
Says PC Magazine:



- A 1970 *Mustang* would do *150 mph*
- A 1971 *microprocessor* (the 1st) ran at *108 KHz*

If cars had increased in speed as much as processors, a new Mustang would do about 22,000,000 mph!

Why Use Computers?



What Is a Computer and What Does It Do?

- Computer: **A programmable, electronic device that accepts data, performs operations on that data, and stores the data or results as needed**
  - Computers follow instructions, called **programs**, which determine the tasks the computer will perform

- 5 Basic Computer operations
  - Input: Entering data into the computer
  - Processing: Performing operations on the data
  - Output: Presenting the results (information)
  - Storage: Saving data, programs, or output for future use
  - Communications: Sending or receiving data

Information is processed data…..

What Can Computers Do?

- Input/Output operations
  - Reads input
  - Writes output
- Processing operations
  - Math (computation)
  - Decision making (logic)



What Is a Computer and What Does It Do?

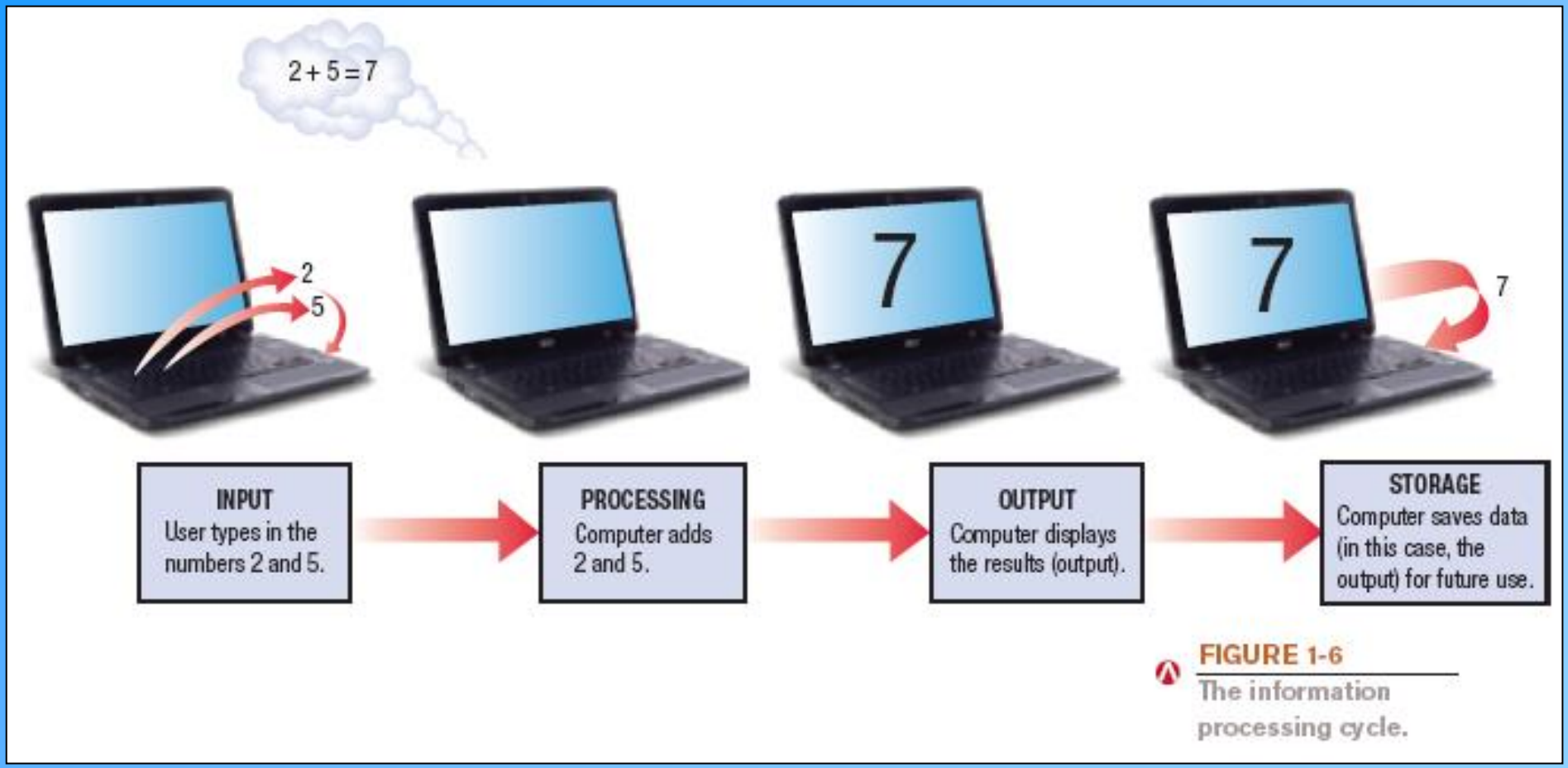
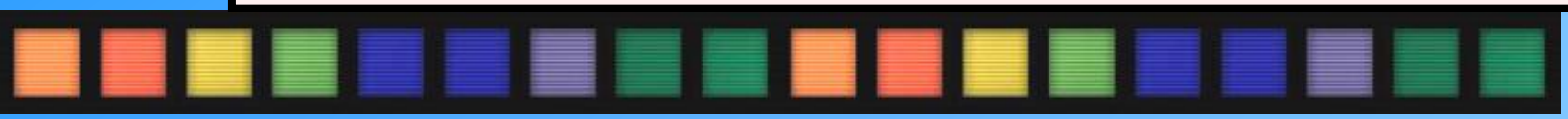


FIGURE 1-6
The information processing cycle.

Why Use Computers?

- Speed
- Accuracy
- Consistency
- Reliability
- Communications
- Memory

Data Representation



Data vs. Information

INPUT



OUTPUT



Data

PROCESS



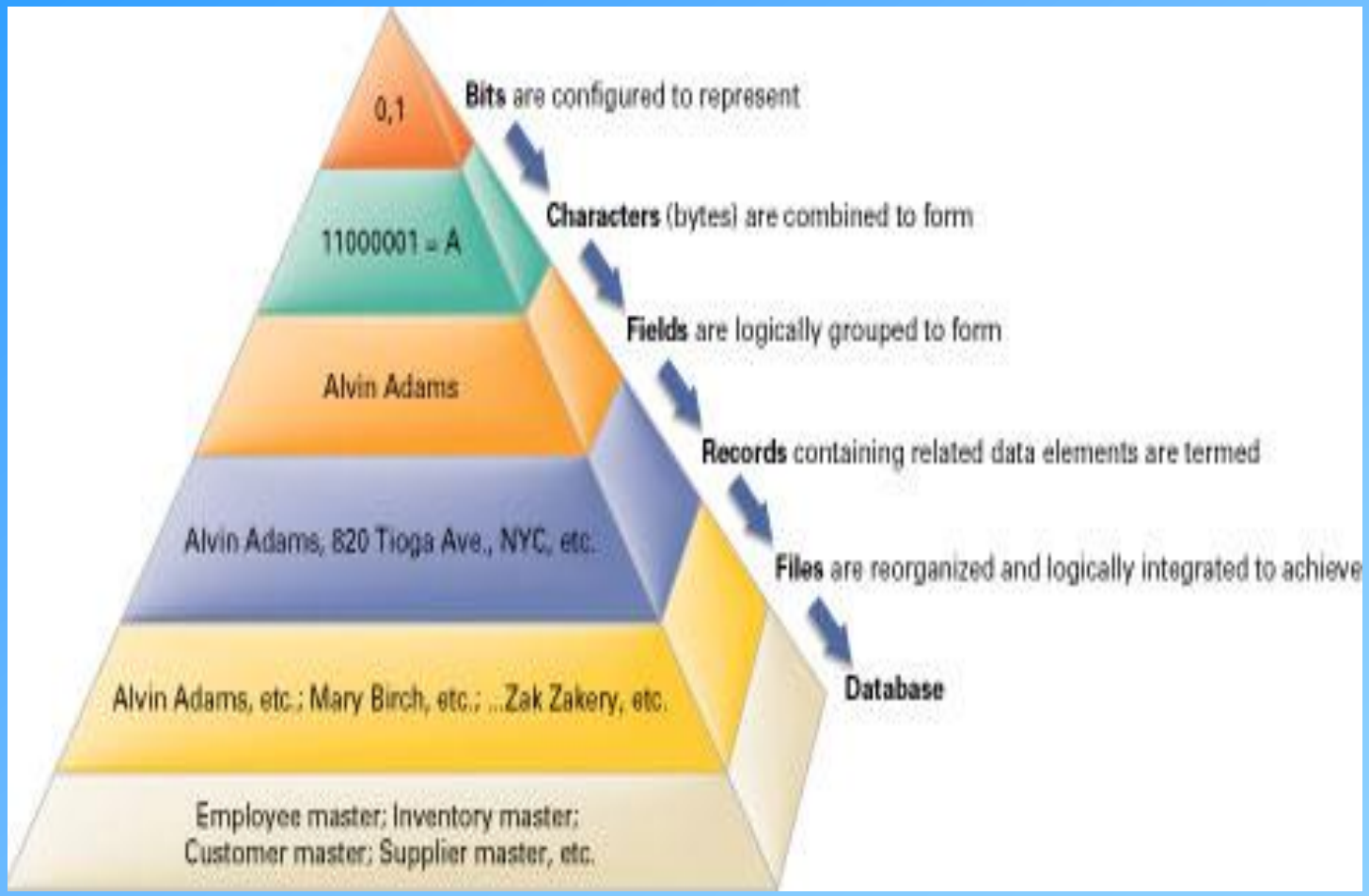
Information

Hierarchy of Data Organization

- Bit: on/off switch
- Character: same as a byte – 8 bits
- Field: group of related characters
- Record: group of related fields
- File: group of related records
- Database: group of related files

Data provide the fuel for the computer system.

Hierarchy of Data Diagram



Using Computers

Personal Computing

- Word Processing
- Spreadsheets
- Database
- Graphics/Presentation
- Desktop Publishing
- Communications
- PIM

Information Systems

- Hardware
- Software
- People
- Procedures
- Data

Communication

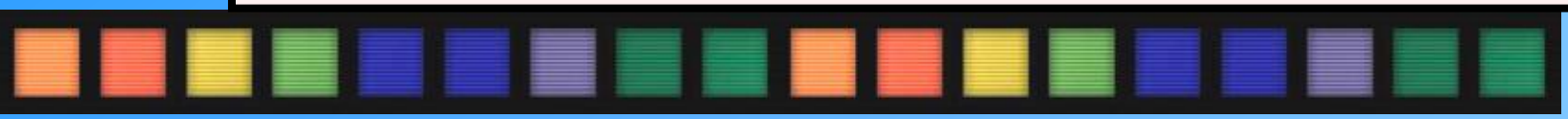
Education & Reference

Entertainment & Edutainment

Science, Research & Engineering



Computers to Fit Every Need



Computers to Fit Every Need

- Handheld Computers
- Notebook PCs
- Desktop PCs
- Thin Client Computers
- Workstations
- Server Computers
- Supercomputers

Computing Power is the key!



Computers to Fit Every Need

- Six basic categories of computers:
  - Embedded computers
  - Mobile devices
  - Personal computers
  - Midrange servers
  - Mainframe computers
  - Supercomputers

Category 1: Embedded Computers

- Embedded computer: Embedded into a product and designed to perform specific tasks or functions for that product
- Cannot be used as general-purpose computers
- Often embedded into:
  - Household appliances
  - Thermostats
  - Sewing machines
  - Cars



FIGURE 1-12
Embedded computers. This car's embedded computers control numerous features, such as notifying the driver when a car enters his or her blind spot.

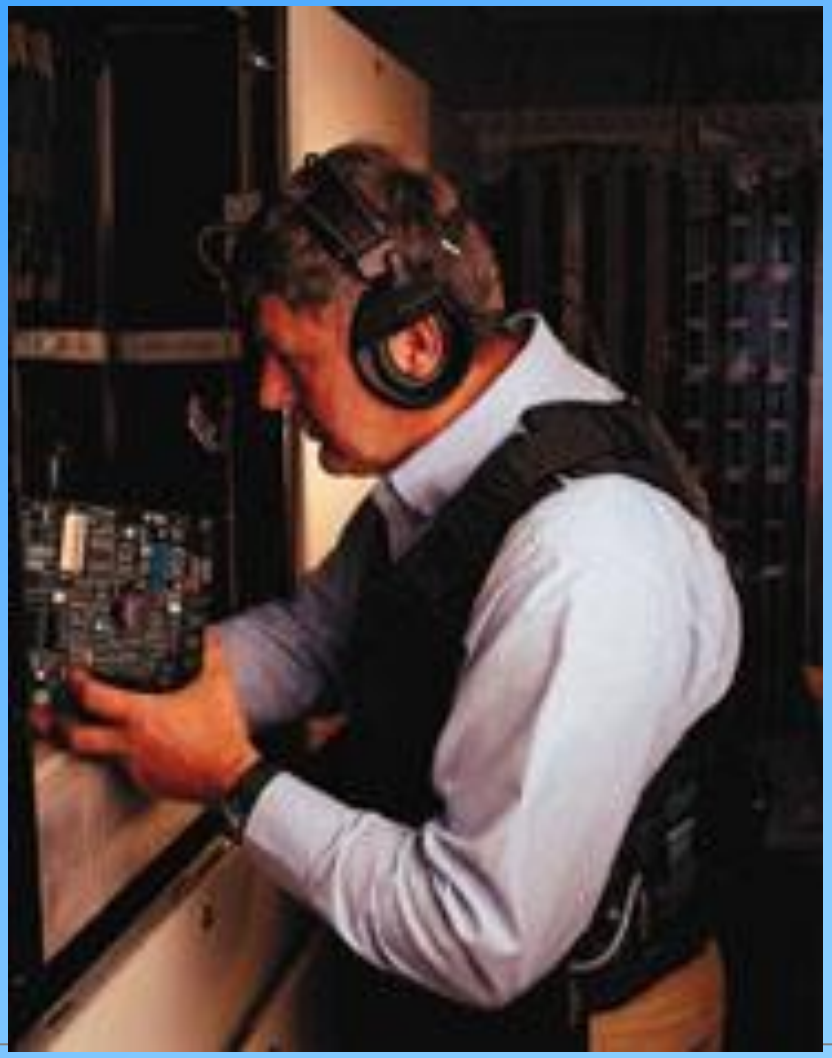
Category 2: Mobile Devices

- Mobile device: A very small device with some type of built-in computing or Internet capability
- Typically based on mobile phones
- Typically have small screens and keyboards
- Examples:
  - Smartphones
  - Handheld gaming devices
  - Portable digital media players



FIGURE 1-13
Smartphones. Most mobile devices today are based on the mobile phone.

Wearable PCs



Handheld Computers

- Palmtop PC
- Personal digital assistants (PDAs)
- Pen-based computers (UPS)
- Connected organizers
- Personal communicators
- Mobile business centers
- Web phones
- Radio Frequency devices (warehouses)



Category 3: Personal Computers (PCs)

- Personal computer: A small computer designed to be used by one person at a time
- Desktop computers: Fit on or next to a desk
  - Can use tower case, desktop case, or all-in-one
  - Can be PC- compatible or Macintosh
  - Not designed to be portable



Category 3: Personal Computers (PCs)…..

IBM PC introduced in 1981

Wintel PC Platform
= Windows 9x/Me/2000
+ Intel (or compatible) processor chip

Mac Platform
= Mac® OS X
+ Motorola ® PowerPC® chip



Category 3: Personal Computers (PCs)…..

- Notebook (laptop) computers: Typically use clamshell design
- Tablet computers: Can be slate tablets or convertible tablets
- Netbooks: Small notebooks; rapidly growing type of PC
- Ultra-mobile PCs (UMPCs): Handheld computers



FIGURE 1-15
Portable computers.

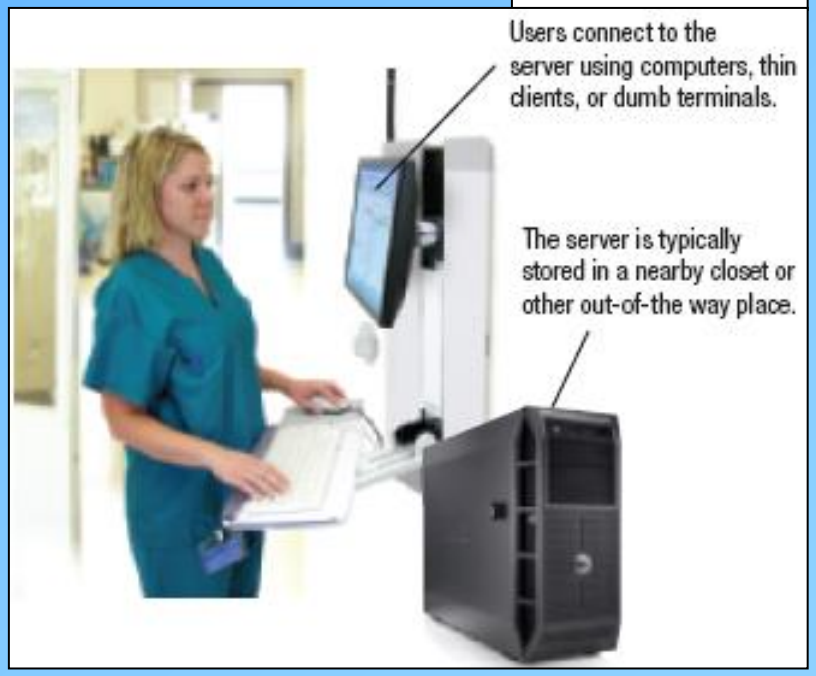


Convertible Tablet

Category 4: Midrange Servers

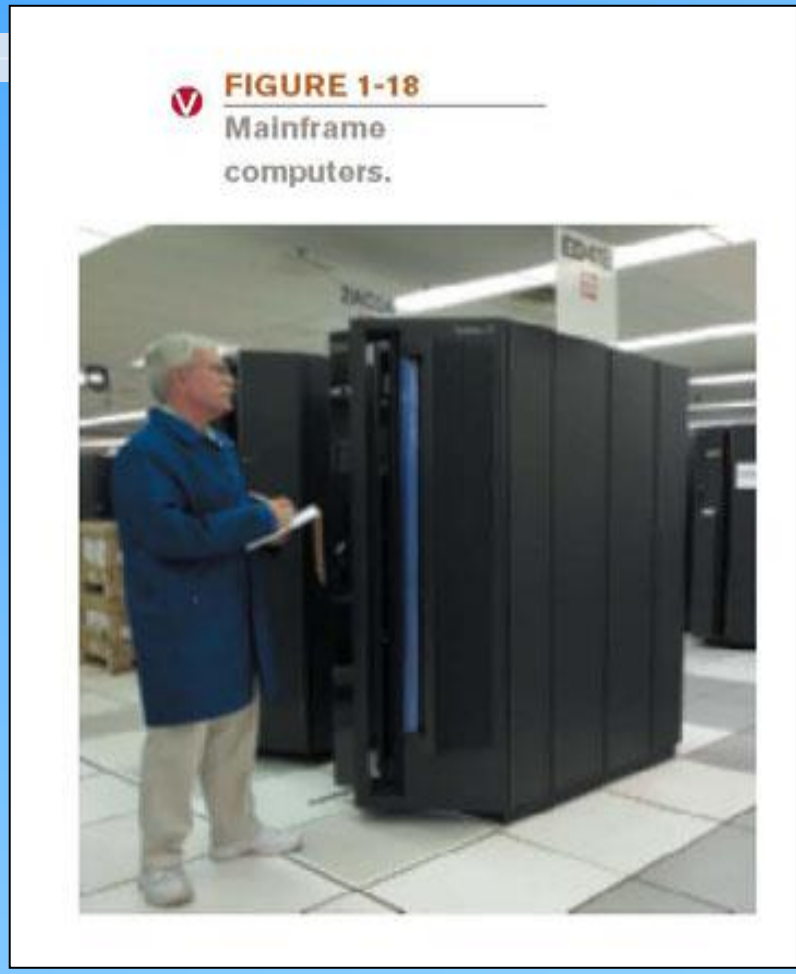
- Midrange server*:* A **medium-sized** computer used to host programs and data for a small network
  - Users connect via a network with a computer or dumb terminal

FIGURE 1-17
Midrange servers. Midrange servers are used to host data and programs on a small network, such as a school computer lab or medical office network.



Category 5: Mainframe Computers

- Mainframe computer: Powerful computer used by several large organizations to manage large amounts of centralized data
  - Standard choice for large organizations, hospitals, universities, large businesses, banks, government offices
  - Located in climate-controlled data centers and connected to the rest of the company computers via a network
  - Larger, more expensive, and more powerful than midrange servers
  - Usually operate 24 hours a day
  - Also called high-end servers or enterprise-class servers


FIGURE 1-18
Mainframe computers.

Category 6: Supercomputers

- Supercomputer: Fastest, most expensive, most powerful type of computer
  - Generally run one program at a time, as fast as possible
  - Commonly built by connecting hundreds of smaller computers, supercomputing cluster
  - Used for space exploration, missile guidance, satellites, weather forecast, oil exploration, scientific research, complex Web sites, decision support systems, 3D applications, etc.



FIGURE 1-19
The Roadrunner supercomputer. Supercomputers are used for specialized situations in which immense processing speed is required.

Category 6: Supercomputers

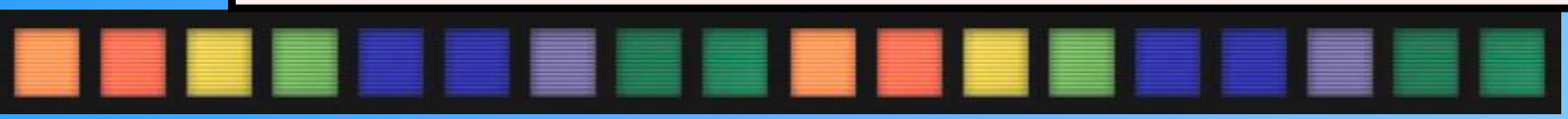
- Processor-bound applications
- 100 times faster than typical server
- Used for
  - Scientific simulations and forecasting
  - Medicine
  - Advanced graphics



Quick Quiz

1. True or False: Microcomputers were invented in the early 80's?
2. True or False: The computer performs 4 basic operations, input, processing, output, and storage
3. True or False: The terms mainframe computer and supercomputer are interchangeable; both refer to the largest, most powerful computers.

Computer Hardware

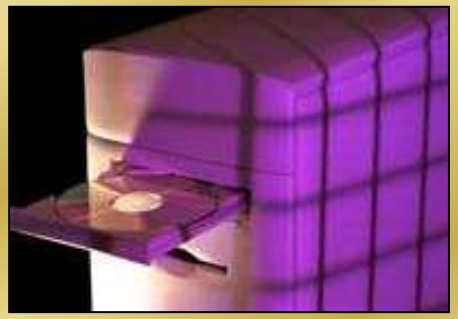


Computer Hardware

Computer System Components



INPUT



STORAGE

Peripherals



PROCESS



OUTPUT

RAM

Chips

Hardcopy

Softcopy

Desktop PCs



- Configuration
  - Motherboard
  - Keyboard
  - Point-and-draw device
  - Monitor
  - Printer
  - Hard disk
  - Floppy disk drive
  - Optical disk drive
  - Microphone
  - Speakers



FLASH MEMORY CARD READER
Reads and writes flash memory cards.

DVD DRIVE
Reads and writes CD and DVD discs.

HARD DRIVE
Located inside the system unit; stores programs and most data.

SYSTEM UNIT
Case that contains the CPU, memory, power supply, storage devices, and all other internal hardware.

MONITOR
Lets you see your work as you go; the primary output device.

PRINTER
Produces printed copies of computer output.

MICROPHONE
Captures spoken input.

SPEAKERS
Produce audio output.

USB PORTS
Connect external devices that use the USB interface.

ROUTER
Connects devices so they can share an Internet connection and data.

MODEM
Connects the computer to the Internet.

KEYBOARD
Used to type instructions into the computer; a primary input device.

CD AND DVD DISCS
Used to deliver programs and store large multimedia files.

MOUSE
Used to make on-screen selections; a primary input device.

FLASH MEMORY CARDS
Used to store digital photos, music files, and other content.

USB FLASH DRIVE
Used to store documents, digital photos, music files, and other content to be moved from one PC to another.

Computer Hardware

- Hardware: The physical parts of a computer
  - **Internal hardware**
    - Located inside the main box (system unit) of the computer
  - **External hardware**
    - Located outside the system unit
    - Connect to the computer via a wired or wireless connection
  - There is hardware associated with all five computer operations

Computer Hardware

- **Input devices**
  - Used to input data into the computer
  - Keyboards, mice, scanners, cameras, microphones, joysticks, touch pads, touch screens, fingerprint readers, etc.
- **Processing devices**
  - Perform calculations and control computer's operation
  - Central processing unit (CPU) and memory
- **Output devices**
  - Present results to the user
  - Monitors, printers, speakers, projectors, etc.

Computer Hardware

- **Storage devices**
  - Used to store data on or access data from storage media
  - Hard drives, CD/DVD discs and drives, USB flash drives, etc.
- **Communications devices**
  - Allow users to communicate with others and to electronically access remote information
  - Modems, network adapters, etc.

Quick Quiz

1. A tablet PC is an example of a(n) _____.
   a. Desktop computer
   b. Portable PC
   c. Internet appliance
2. Speakers are an example of a(n) _____ device.
3. True or False: A smartphone is an example of a mobile device

Computer Software

Computer Software

- Programs

System Software

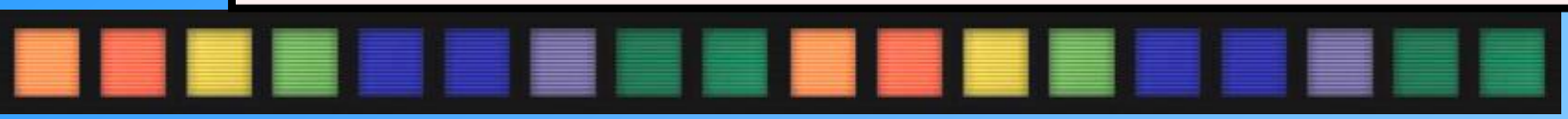


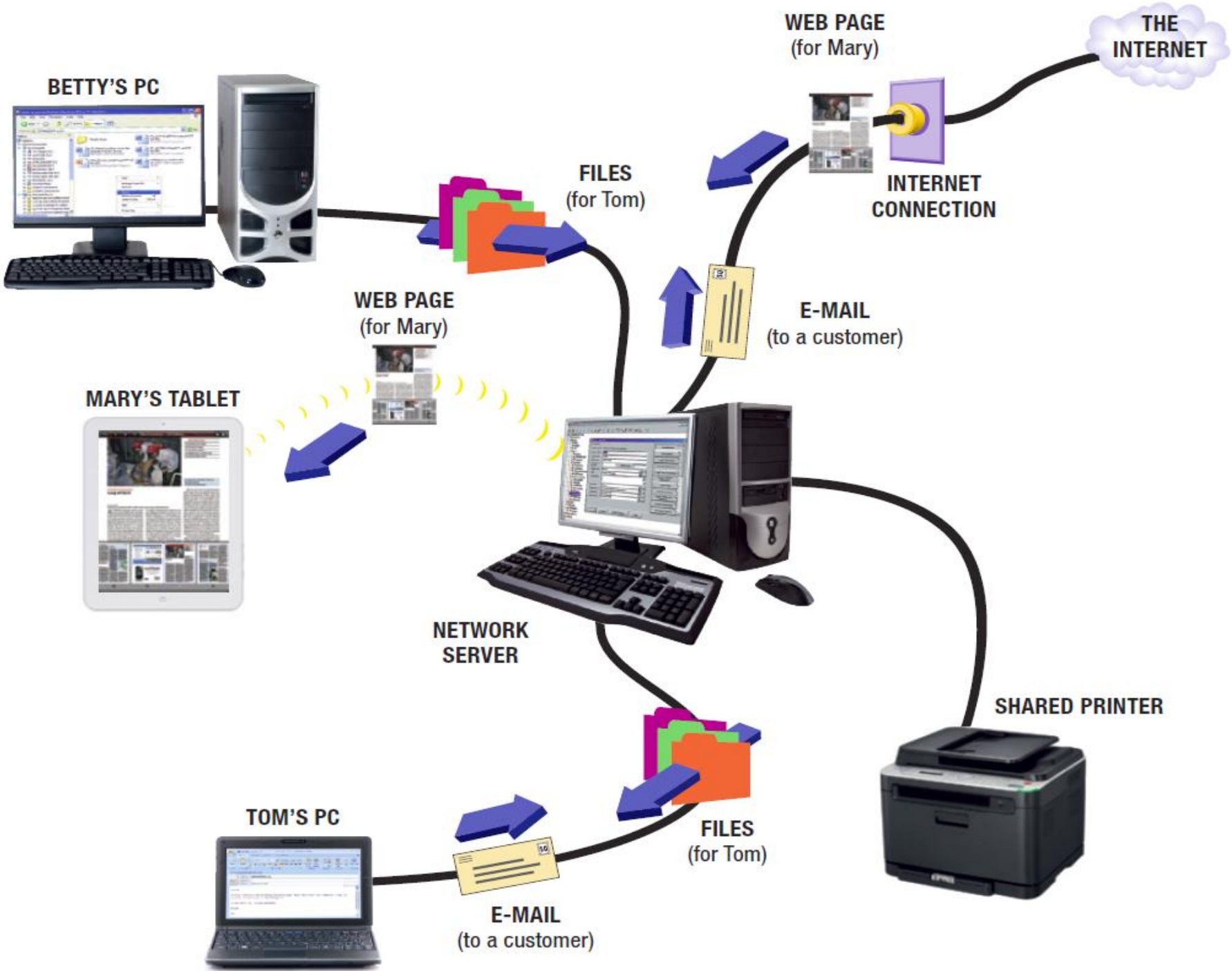
Manage, maintain & control system resources

Computer Software

- **System Software**
  - The Operating System
  - Support/Service Software
- **Application Software**
  - Allow users to perform their specific tasks
  - Ready-Made Software
  - Write your own software application programs!!!
  - What do you need to know?

Computer Networks




WEB PAGE
(for Mary)
THE
INTERNET
BETTY'S PC
FILES
(for Tom)
INTERNET
CONNECTION
WEB PAGE
(for Mary)
E-MAIL
(to a customer)
MARY'S TABLET
NETWORK
SERVER
SHARED PRINTER
TOM'S PC
FILES
(for Tom)
E-MAIL
(to a customer)

Going Online

- Internet (*the Net*)
- Internet Service Provider (ISP)
- Information Service (like AOL)
- Modems
- Online/Offline
- Download/Upload



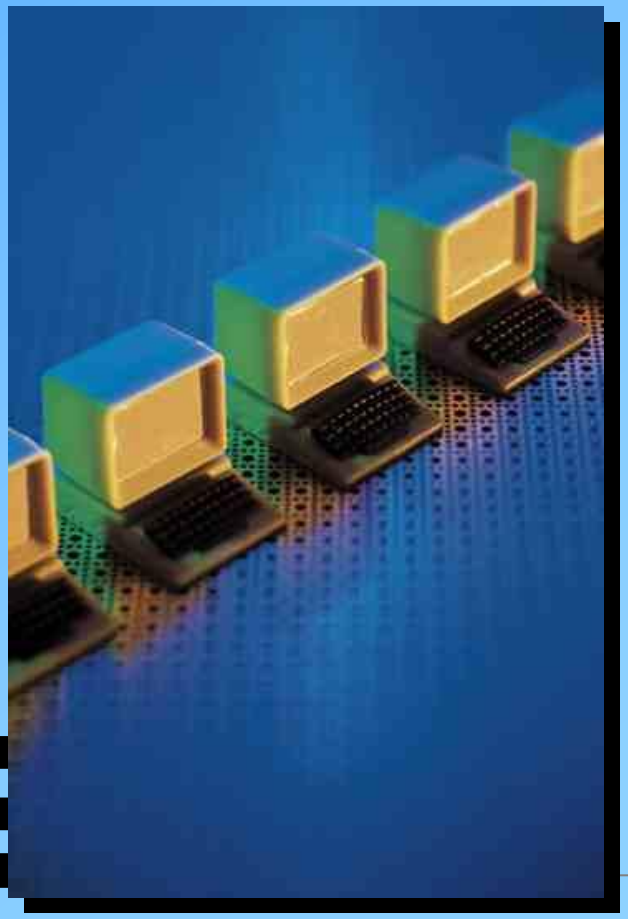
Internet Services



- E-mail
- Chat
- Videophone
- Ftp
- Newsgroups
- World Wide Web

How Thin Clients Differ From PCs

- Smaller processor
- Less RAM
- No hard disk
- Less expensive
- Depends on server



Client/Server Computing

- Server
  - PC to Supercomputer
  - Data storage and applications software
- Client
  - PC, workstation or thin client
  - Requests processing or support from server
- Applications Software
  - Front-end (client)
  - Back-end (server)
- Enterprise-wide Systems
- Proxy server computer

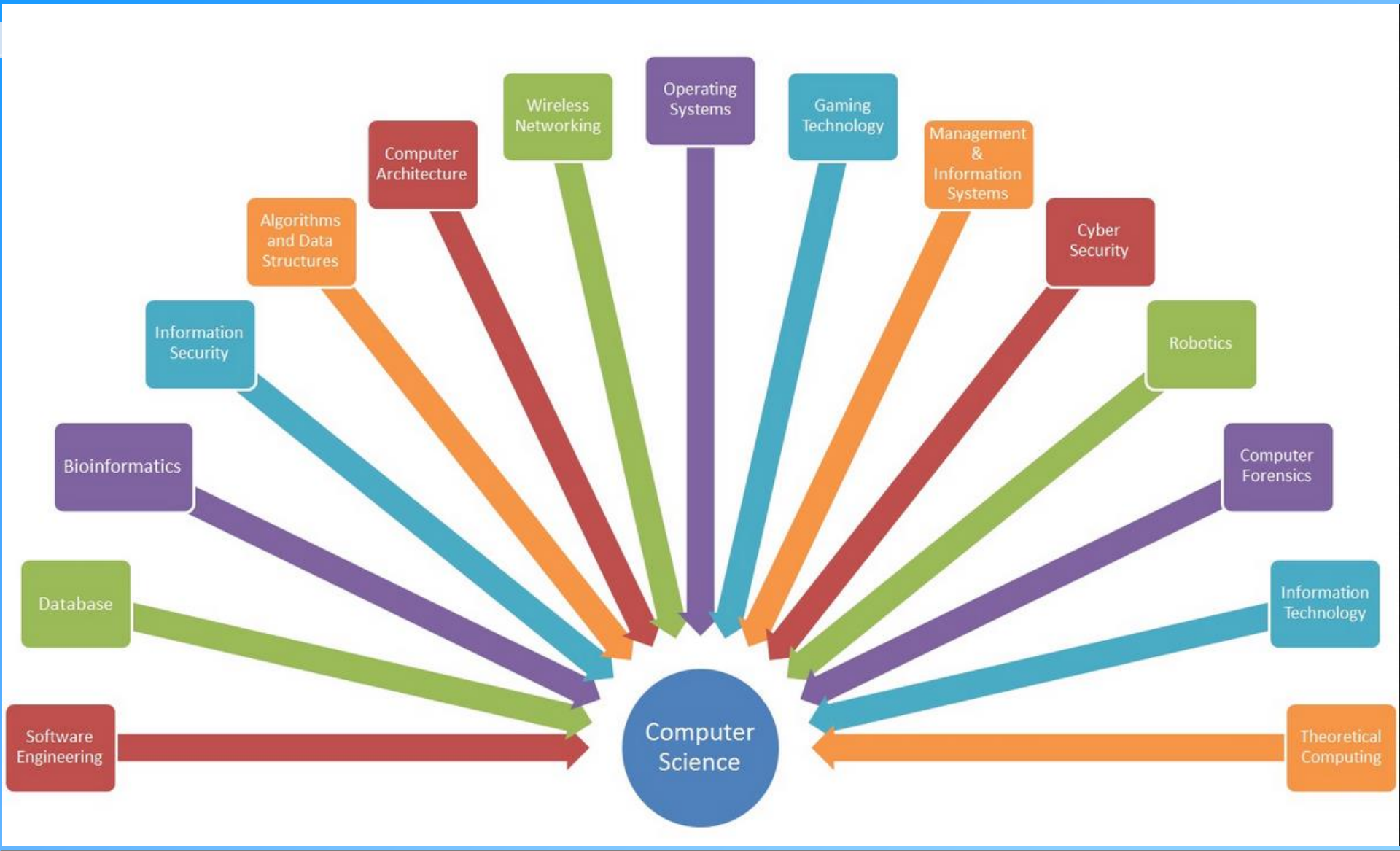


Fields of Study

AI Fields

- **Intelligent Systems**
  - Conventional Systems
  - Machine Learning
  - Artificial Neural Networks (ANN)
  - Deep Learning
- **Data Science Software**
  - Big Data
  - Data processing/Distributed Data
  - Data Visualization
  - Artificial Neural Networks (ANN)
  - Shallow/Deep Learning
  - Internet Of Things (IOT)

Computer Science Fields



MAP OF COMPUTER SCIENCE

THEORETICAL COMPUTER SCIENCE

COMPUTATIONAL COMPLEXITY
DOES P≠NP?
NP-COMPLETE
NP
P
EFFICIENT FOR A QUANTUM COMPUTER
EFFICIENT FOR A COMPUTER

INFORMATION THEORY
ENTROPY
COMPRESSION
ERROR CORRECTION
PARITY CHECKING
CODING THEORY

CRYPTOGRAPHY
PUBLIC KEY
PRIVATE KEY
ALL THE SECRETS

COMPUTABILITY THEORY
ALAN TURING
LAMBDA CALCULUS
COMPUTATIONAL GEOMETRY
GRAPH THEORY
AUTOMATA THEORY
QUANTUM COMPUTATION

TURING MACHINE
STATE REGISTER
INFINITELY LONG TAPE
HEAD
1: MOVE LEFT
2: MOVE RIGHT
3: FLIP BIT
123: STOP

AND MORE
PARALLEL PROGRAMMING
JOB 1
JOB 2
JOB 3
JOB 4
MAIN JOB
SMALLER JOBS

ALGORITHMS
BUBBLE SORT(A):
1: GO FROM LEFT TO RIGHT
2: COMPARE EACH PAIR
3: IF LEFT ONE HIGHER, SWITCH
4: DO UNTIL NO MORE SWITCHES
BUBBLE SORT O(n²)
MERGE SORT O(n log n)
ANALYSIS OF ALGORITHMS
ALGORITHMIC COMPLEXITY

LOGIC
ZERO
ONE
NOT
OR
NOR
AND
NAND
XOR

DATA STRUCTURES
ARRAY
LINKED LIST
TREE
GRAPH
STACK
HASHING
FORMAT METHODS

COMPUTER ENGINEERING

COMPUTER ARCHITECTURE
CPU
CENTRAL PROCESSING UNIT
CONTROL UNIT
ARITHMETIC/LOGIC UNIT
MEMORY UNIT
INPUT
OUTPUT
GPU
MULTIPROCESSORS
FPGA
LOGIC BLOCK
INTERCONNECTION
INPUT/OUTPUT
SWITCH BOX
CONNECT BLOCK

SCHEDULING
PROCESSES
SCHEDULER
CPU 1
CPU 2
CPU 3
CPU 4
MULTIPROCESSING

HARDWARE
MONITOR
SOUND
SSD
GPU
RAM
POWER
INPUT
CPU
MOTHERBOARD

SOFTWARE AND PROGRAMMING LANGUAGES
PYTHON
JAVASCRIPT
PERL
PHP
SWIFT
SQL
JAVA
C#
C++
C
ASSEMBLY
WEB APPS
BROWSER
APPLICATIONS
OPERATING SYSTEM
BIOS
SILICON
MACHINE CODE
JAVASCRIPT
COMPILER
C++
COMPILER
C
COMPILER
ASSEMBLY
COMPILER

COMPILERS

DATA MANAGEMENT
DATABASES
SQL
DATACENTRES

NETWORKING
CONCURRENT/DISTRIBUTED/PARALLEL SYSTEMS

OPERATING SYSTEMS
iOS
macOS

SOFTWARE ENGINEERING
FORMAL METHODS
UNIT TESTING
VERSION CONTROL
OBJECT ORIENTED DESIGN

PERFORMANCE
COMPUTER ANALYSIS
BENCHMARKING

APPLICATIONS

MACHINE LEARNING
SUPERVISED
UNSUPERVISED
REINFORCEMENT
NEURAL NETWORK
CAT

OPTIMISATION
FINANCE
LEAGUE OF LEGENDS
AMAZON WAREHOUSE

BOOLEAN SATISFIABILITY
x_1 OR x_2 OR $\bar{x}_3$ (SAT)
$\bar{x}_1$ OR $\bar{x}_2$ OR x_3
$\bar{x}_1$ OR x_1 OR $\bar{x}_3$
$\bar{x}_1$ OR x_1 OR x_3

SUPER COMPUTING

COMPUTER GRAPHICS

COMPUTATIONAL SCIENCE
COMPUTATIONAL PHYSICS
NUMERICAL ANALYSIS
BIOINFORMATICS
COMPUTATIONAL CHEMISTRY

HACKING

ARTIFICIAL INTELLIGENCE
A.I.

ROBOTICS

COMPUTER VISION
FIND THE HUMANS

VIRTUAL REALITY

SIMULATION

BIG DATA

NATURAL LANGUAGE PROCESSING
CHATBOTS
-HEY
-ARE U A ROBOT?
-YES
-WAIT, I MEAN NO
-PROVE IT!
-CALCULATING...
-I MADE YOU CAKE-
SCONES
TEA
FLOUR
BUTTER
BIRTHDAY
CAKE
FOOD
BACON
CELEBRATION
PANCAKE
BREAKFAST
KNOWLEDGE REPRESENTATION

TELEPRESCENCE

AUGMENTED REALITY
YOUTUBE
DOMAIN OF SCIENCE
MAP OF COMPUTER SCIENCE

HUMAN COMPUTER INTERACTION

IMAGE PROCESSING

INTERNET OF THINGS

BY DOMINIC WALLIMAN ©2017

Review

- Define COMPUTER
- What are the 6 categories of computers
- What are the 5 basic operations
- What is hardware?
- What hardware comprises:
  - Input devices
  - Processing devices
  - Output Devices
  - Storage devices
  - Communication devices?
- What Software comprises:
  - SystemApplication Software